Plotly

In [17]:
```
import plotly.express as px
```

In [18]:
```
data = px.data.gapminder()
data.head()
```

Out[18]:

	country	continent	year	lifeExp	pop	gdpPercap	iso_alpha	iso_num
0	Afghanistan	Asia	1952	28.801	8425333	779.445314	AFG	4
1	Afghanistan	Asia	1957	30.332	9240934	820.853030	AFG	4
2	Afghanistan	Asia	1962	31.997	10267083	853.100710	AFG	4
3	Afghanistan	Asia	1967	34.020	11537966	836.197138	AFG	4
4	Afghanistan	Asia	1972	36.088	13079460	739.981106	AFG	4

Task 1. Create a line chart shwoinng how life expectancy has changed over time for India and Canada.

Modify the chart so that:

1. Lines are drawn using spline interpolation instead of straight segments.
2. Increase the smoothing level of the spline.
3. Change the line style to 'longdashdot'.
4. Increase the line width to make trends more visible.
5. Change the line color manually instead of using default color mapping.

In [19]:
```
data_fil = data[data['country'].isin(['India', 'Canada'])]
fig = px.line(data_fil, x='year', y='lifeExp', color='country', markers=True, title
fig.update_traces(line_shape='spline', selector=dict(name='India'))
fig.update_traces(line_smoothing=0.7, selector=dict(name='India'))
fig.update_traces(line_dash='longdashdot', selector=dict(name='Canada'))
fig.update_traces(line_width=3, selector=dict(type='scatter'))
fig.update_traces(line_color='red', selector=dict(name='India'))
```

Task 2. Create a scatter plot to explore the relationship between GDP per capita and lifeexpectancy across continents.

1. Make the marker symbol unique for each continent instead of the default circle.
2. Add borders to the markers for 'Asia' with black border color, border width of 2 pixels.
3. Reduce the opacity of other markers.
4. Convert the GDP per capita axis to a logarithmic scale to better visualize the spread of values.
5. Customize the hover information to display:

- Country name, GDP per capita, Life expectancy, Population

1. Add animation so that the visualization changes across years. Change background color of animation.

In [20]:
```
fig = px.scatter(data_fil, x = 'gdpPercap', y = 'lifeExp', color='continent')
fig.show()
```

In [21]:
```
fig.update_traces(selector=dict(name='Asia'), marker=dict(line=dict(color='black',
```

In [22]:
```
fig.update_traces(marker=dict(opacity=0.4))
```

In [23]:
```
fig.update_xaxes(type='log')
```

In [24]:
```
fig.update_traces(hoverinfo='all')
```

In [25]:

```
fig.update_layout( sliders=[dict(
        bgcolor="blue",
        activebgcolor="red",
        transition=dict(duration=1500)
    )])
```

Task 3. Create a box plot to compare the distribution of life expectancy across different continents. Use data from the year 2007

In [26]:
```
data_2007 = data_fil[data_fil["year"] == 2007]
print(data_2007.head())
fig = px.box(
    data_2007,
    x="continent",
    y="lifeExp",
    color="continent",
    title="Life Expectancy Distribution by Continent (2007)",
    points="all"
)
fig.show()
```

```
    country continent  year  lifeExp         pop     gdpPercap iso_alpha  \
251  Canada  Americas  2007   80.653    33390141  36319.235010       CAN
707   India      Asia  2007   64.698  1110396331   2452.210407       IND

     iso_num
251      124
707      356
```

3.1 Display Individual Data Points (Show all individual country data points along with the box plot).

In [27]:
```
fig.update_traces(boxpoints='all', jitter=0.35, pointpos=0)
fig.show()
```

3.2. Display the Mean Value (Add a mean line inside each box to highlight the average life expectancy).

In [28]:
```
fig.update_traces(boxmean=True)
fig.show()
```

3.3. Enable Notches in the Box Plot (Helps to indicate confidence around the median).

In [29]:
```
fig.update_traces(notched=True)
fig.show()
```

3.4 Improve Layout by adding an appropriate title, adjusting box width and rotating the x-axis labels slightly.

In [30]:
```
fig.update_layout(title=dict(text="Life expectancy across 2007 by continents"), xax
fig.show()
```

Task 4. Create separate scatter plots using faceting to compare the relationship between GDP per capita and life expectancy across continents

In [31]:
```
fig = px.scatter(data,
                 x='gdpPercap',
                 y='lifeExp',
                 color='continent',
                 facet_col='continent',
                 animation_frame='year',
                 animation_group='country',
                 hover_name='country',
                 log_x=True,
                 size_max=60,
                 title='GDP per Capita vs. Life Expectancy (1952-2007) by Continent
fig.show()
```

4.1. Apply Log Scale to GDP Axis so that country distributions are easier to observe.

In [32]:
```
fig.update_traces(marker_line_width=1, marker_line_color='black')
fig.show()
```

4.2. Add Marker Borders so that overlapping points are easier to distinguish.

In [33]:
```
fig.update_traces(marker=dict(line=dict(color="black", width=1)))
fig.show()
```

4.3. Adjust Spacing Between Facets.

In [34]:
```
fig.update_layout(
    grid=dict(
        xgap=0.12,
        ygap=0.12
    )
)
fig.show()
```

4.4. Modify the subplot titles so they clearly indicate the continent name.

In [35]:
```
fig.for_each_annotation(lambda a: a.update(text=f"Continent: {a.text.split('=')}"[-
fig.show()
```

4.5. Improve Layout by adjusting the figure size and positioning the legend appropriately.

In [36]:
```
fig.update_layout(
    width=1000,
    height=500,
    showlegend=True,
    legend=dict(
        title="Continent",
        orientation="h",
        yanchor="bottom",
        y=1.02,
        xanchor="center",
        x=0.5
    ),
    margin=dict(t=100, b=30, l=30, r=10)
)
fig.show()
```

Task 5. Create a histogram to study the distribution of life expectancy across countries. Use data from the year 2007.

In [37]:
```
data_2007 = data[data["year"] == 2007]
fig = px.histogram(
    data_2007,
    x="lifeExp",
    nbins=20,
    title="Distribution of Life Expectancy (2007)",
    labels={"lifeExp": "Life Expectancy"}
 )
fig.show()
```

5.1 Adjust the Number of Bins

In [38]:
```
fig.update_traces(nbinsx=30)
fig.show()
```

5.2 Overlay the Histograms (overlapping each other instead of stacking).

In [39]:
```
fig = px.histogram(
    data_2007,
    x="lifeExp",
    color="continent",
    nbins=30,
    barmode="overlay",
    opacity=0.6,
    title="Distribution of Life Expectancy (2007) - Overlaid by Continent",
    labels={"lifeExp": "Life Expectancy"}
)
fig.show()
```

5.3. Add a Marginal Plot (Include a box plot or violin plot along the axis).

In [40]:
```
fig = px.histogram(
data_2007,
x="lifeExp",
color="continent",
nbins=30,
barmode="overlay",
opacity=0.6,
marginal="box",
title="Distribution of Life Expectancy (2007) with Marginal Box Plot",
labels={"lifeExp": "Life Expectancy"}
)
fig.show()
```

5.4. Show Probability Distribution (Normalize the histogram so the y-axis represents probability instead of count).

In [41]:
```
fig.update_traces(histnorm="probability", selector=dict(type="histogram"))
fig.update_layout(yaxis_title="Probability")
fig.show()
```

5.5. Improve Layout by adjusting bar opacity and improving legend placement

In [42]:
```
fig.update_traces(opacity=0.45, selector=dict(type="histogram"))
fig.update_layout(
    legend=dict(
      title="Continent",
      orientation="h",
      yanchor="bottom",
      y=1.02,
      xanchor="center",
      x=0.5
    )
)
fig.show()
```

In [42]: